Koneru Lakshmaiah Education Foundation
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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.
Phone No: 7815926816, www.klh.edu.in

Department of CSE

B.Tech. CSE Trimester - 5

Roll Number	2520030363
Name	G.Akshay

Cloud Infrastructure and Services

25CS2206E

LAB RECORD

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CDIS – Y25

Experiment - 1

AIM: To create an AWS free tier account to provision the resources, create an IAM user with programmatic access, generate access keys, and configure AWS Free Tier and billing alerts.

Description:

This experiment covers the creation of an AWS Identity and Access Management (IAM) user with programmatic access only, generation and secure download of the Access Key ID and Secret Access Key, enabling AWS Free Tier usage notifications, and configuring an AWS Budget with email alerts. These steps help monitor AWS usage and control unexpected costs.

Prerequisites

• Active AWS account.

• Access to the AWS Management Console using the root user or an account with administrative permissions.

• Valid email ID for receiving billing notifications.

Steps:

Part A: Create an IAM User with Programmatic Access Only

1. Log in to the AWS Management Console using your root account credentials.
2. In the search bar at the top of the AWS Console, search for IAM and select Identity and Access Management (IAM).
3. From the left navigation panel, select Users.
4. Click **Create user**.
5. Enter a suitable username (e.g., Student-IAM-User).

6. Do not enable AWS Management Console access, as the user requires programmatic access only.
7. Click **Next**.
8. Under Set permissions, select **Attach policies directly**.
9. Select the required permissions policy based on the tasks to be performed. For example, select Administrative access based on the tasks.
10. Click Next and review the configuration.
11. Click Create user to complete the process.
12. Verify that the newly created IAM user appears in the Users list.

Part B: Generate and Download Access Keys

1. In the IAM dashboard, select the newly created IAM user.
2. Open the Security credentials tab.
3. Scroll down to the Access keys section.
4. Click **Create access key**.
5. Select the appropriate use case, such as Command Line Interface (CLI), if the key is intended for AWS CLI access.
6. Acknowledge the recommendation and click Next.
7. Optionally enter a description tag (e.g., Student-CLI-Access-Key).
8. Click Create access key.
9. The console displays the Access Key ID and Secret Access Key.
10. Click Download .csv file to download the credentials securely.
11. Verify that the downloaded CSV file contains both the Access Key ID and Secret Access Key.
12. Click Done.

Experiment - 1 (continued)

Part C: Configure AWS Free Tier Usage Alerts

Configure alerts through Billing Preferences.

1. Log in to the AWS Management Console using the Root User account.
2. Search for Billing and Cost Management in the AWS search bar and open it.
3. In the left navigation panel, locate Preferences and select Billing preferences.
4. Locate the Alert preferences section.
5. Click Edit.
6. Select the checkbox **Receive AWS Free Tier usage alerts**.
7. Under Additional email address to receive alerts, enter the email address where notifications should be received (if required).
8. Click Update to save the settings.
9. Verify that the Free Tier usage alert option is enabled.

Part D: Configure AWS Billing Alerts Using the Custom Budget Option

1. Log in to the AWS Management Console using the root account.
2. Search for Billing and Cost Management and open it.
3. From the left navigation panel, select Budgets under Budgets and Planning.
4. Click Create budget.

5. Under Budget setup, select Use a template (simplified). In Templates – new, select Zero spend budget (notifies you once spending exceeds $0.01), then enter the budget name and email ID and go to step 17; or select Customize (advanced) instead of using a template.
6. Under Budget types, select Cost budget.
7. Click Next.
8. Configure the budget details as shown below.

Budget name	AWS-Monthly-Cost-Budget
Period	Monthly
Budget renewal type	Recurring budget
Start month	Current month
Budgeting method	Fixed
Budgeted amount	$0 (example)
Budget scope	All AWS services

9. Keep the advanced settings at their default values unless specific requirements exist.
10. Click Next.
11. Under Configure alerts, click Add an alert threshold.

Configure the alert as follows:

Threshold	0

Threshold type	% of budgeted amount
Alert trigger	Actual
Email recipients	Your registered email address

12. Click Next after configuring the alert thresholds.
13. On the Attach actions page, skip the optional budget actions for this experiment.
14. Click Next.
15. Review all the configured budget details and alert thresholds.
16. Click Create budget.
17. Verify that the newly created budget appears in the AWS Budgets dashboard.
18. Check your email for AWS budget notifications when the configured thresholds are reached. Verify your email address if AWS requests confirmation.

Output:

Expected output after successful completion of the experiment:

• An IAM user is created without AWS Management Console access.

• Programmatic access credentials (Access Key ID and Secret Access Key) are generated and downloaded securely.

• AWS Free Tier usage notifications are enabled.

• A monthly AWS Budget is created with an email alert threshold.

• The AWS Budgets dashboard displays the created budget, and email notifications can be received when configured thresholds are reached.

Note: Do not share the Access Key ID/Secret Access Key or include them in this lab record. The output above describes the expected result; add your own console screenshots if your instructor requires evidence.

